Lecture(7)

Chapter(2)

Set Theory

- Introduction
- Veen diagrams
- Set operations

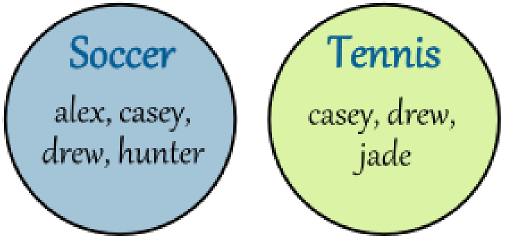
Introduction

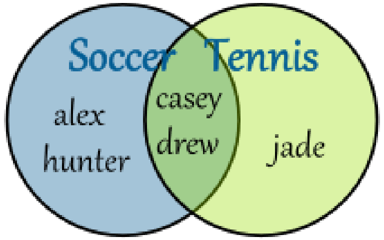
Two, or more, sets can be combined in many different ways. For instance, starting with the set of mathematics majors at your school and the set of computer science majors at your school, we can form the set of students who are mathematics majors or computer science majors, the set of students who are joint majors in mathematics and computer science, the set of all students not majoring in mathematics, and so on.

Set operations

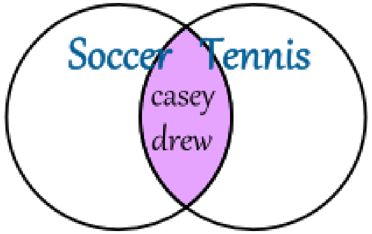
Veen diagram

A Venn diagram is a drawing, in which circular areas represent groups of items usually sharing common properties.





Venn Diagram:
Union of 2 Sets



Venn Diagram:
Intersection of 2 Sets

Set operations

Universal set

It's a set that contains everything.

Well, not exactly everything. Everything that is relevant to our question.

Then our sets included integers. The universal set for that would be all the integers.

Set operations

Union

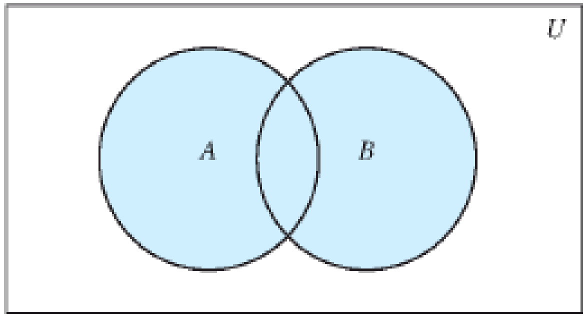
Definition:

Let A and B be sets. The *union* of the sets A and B, denoted by $A \cup B$, is the set that contains those elements that are either in A or in B, or in both.

An element x belongs to the union of the sets A and B if and only if x belongs to A or x belongs to B.

This tells us that $A \cup B = \{x \mid x \in A \vee x \in B\}$.

Set operations



$A \cup B$ is shaded.

Example:

The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$.

Set operations

Intersection

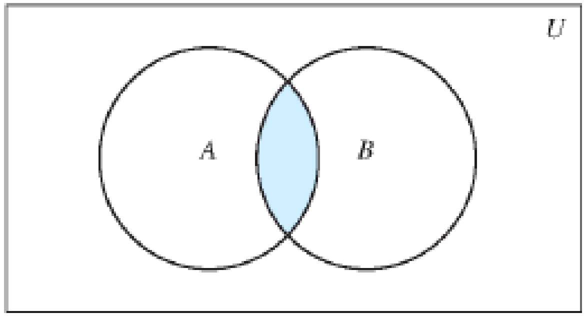
Definition:

Let A and B be sets. The *intersection* of the sets A and B, denoted by $A \cap B$, is the set containing those elements in both A and B.

An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B.

This tells us that $A \cap B = \{x \mid x \in A \wedge x \in B\}$.

Set operations



$A \cap B$ is shaded.

Example:

The intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 3\}$; that is, $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$.

Set operations

Disjoint

Definition:

Two sets are called *disjoint* if their intersection is the empty set.

Example:

Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$. Because $A \cap B = \emptyset$, A and B are disjoint. ◀

Set operations

Difference

Definition:

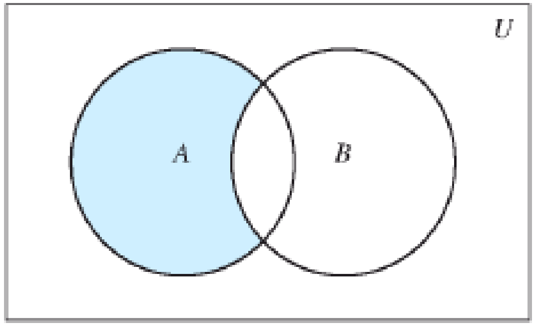
Let A and B be sets. The *difference* of A and B, denoted by $A - B$, is the set containing those elements that are in A but not in B. The difference of A and B is also called the *complement of B with respect to A*.

An element x belongs to the difference of A and B if and only if x belongs to A and x not belongs to B.

This tells us that $A - B = \{x \mid x \in A \wedge x \notin B\}$.

Remark: The difference of sets A and B is sometimes denoted by $A \setminus B$.

Set operations



$A - B$ is shaded.

Example:

The difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{5\}$; that is, $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$. This is different from the difference of $\{1, 2, 3\}$ and $\{1, 3, 5\}$, which is the set $\{2\}$.

Set operations

Complement of a set

Definition:

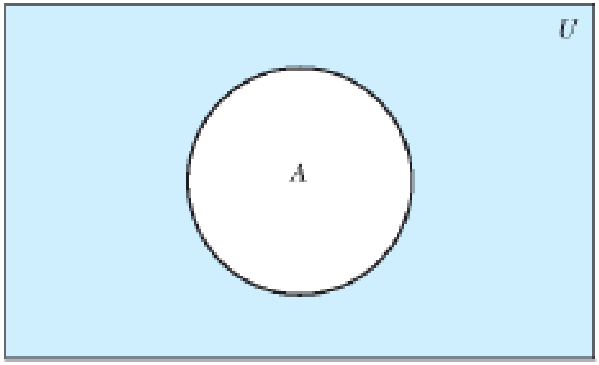
Let U be the universal set. The *complement* of the set A, denoted by $\overline{A}$, is the complement of A with respect to U. Therefore, the complement of the set A is $U - A$.

An element belongs to $\overline{A}$ if and only if $x \notin A$.

This tells us that $\overline{A} = \{x \in U \mid x \notin A\}$.

Remark: the complement of set A is sometimes denoted by A'

Set operations



$\overline{A}$ is shaded.

Example:

Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). Then $\overline{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$.

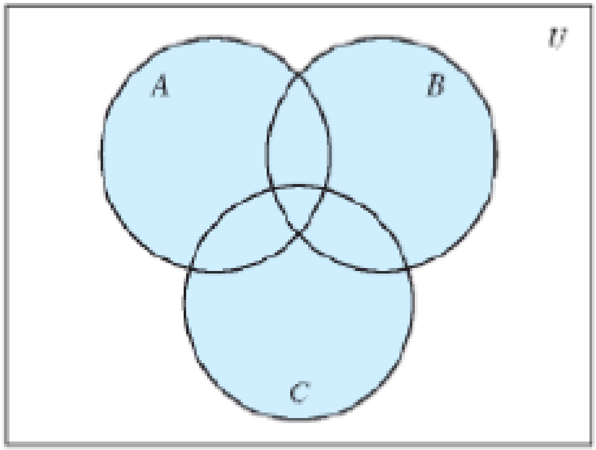
Let A be the set of positive integers greater than 10 (with universal set the set of all positive integers). Then $\overline{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

Set operations

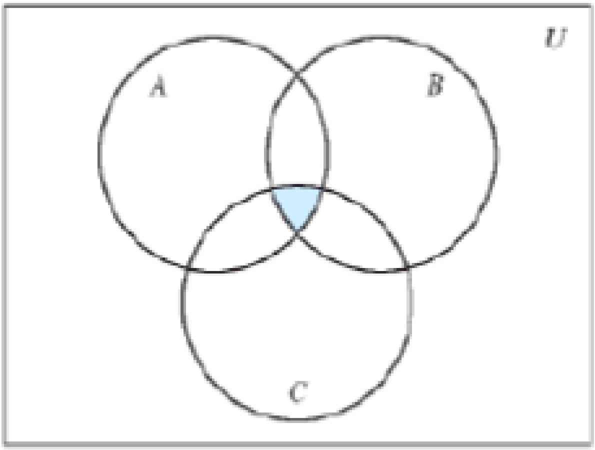
Generalized Unions and Intersections

Suppose A, B, and C are sets. Note that

- $A \cup B \cup C$ contains those elements that are in at least one of the sets A, B, and C.
- $A \cap B \cap C$ contains those elements that are in all of A, B, and C.
- These combinations of the three sets, A, B, and C, are shown below



(a) $A \cup B \cup C$ is shaded.



(b) $A \cap B \cap C$ is shaded.

Set operations

Union of a collection

Definition:

The *union* of a collection of sets is the set that contains those elements that are members of at least one set in the collection.

We use the notation $A_1 \cup A_2 \cup \cdots \cup A_n = \bigcup_{i=1}^{n} A_i$

to denote the union of the sets $A_1, A_2, \ldots, A_n$.

Example: For $i = 1, 2, \ldots$, let $A_i = \{i, i+1, i+2, \ldots\}$. Then,

$$\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{i, i+1, i+2, \ldots\} = \{1, 2, 3, \ldots\},$$

Set operations

Intersection of a collection

Definition:

The *intersection* of a collection of sets is the set that contains those elements that are members of all the sets in the collection.

We use the notation $A_1 \cap A_2 \cap \cdots \cap A_n = \bigcap_{i=1}^{n} A_i$

to denote the union of the sets $A_1, A_2, \ldots, A_n$.

Example: For $i = 1, 2, \ldots$, let $A_i = \{i, i+1, i+2, \ldots\}$. Then,

$$\bigcap_{i=1}^{n} A_i = \bigcap_{i=1}^{n} \{i, i+1, i+2, \ldots\} = \{n, n+1, n+2, \ldots\} = A_n.$$

Exercises

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{0, 3, 6\}$. Find

a) $A \cup B$. **b)** $A \cap B$.
c) $A - B$. **d)** $B - A$.

Let $A = \{0, 2, 4, 6, 8, 10\}$, $B = \{0, 1, 2, 3, 4, 5, 6\}$, and $C = \{4, 5, 6, 7, 8, 9, 10\}$. Find

a) $A \cap B \cap C$. **b)** $A \cup B \cup C$.
c) $(A \cup B) \cap C$. **d)** $(A \cap B) \cup C$.

Draw the Venn diagrams for each of these combination of the sets A, B, and C.

a) $A \cap (B - C)$ **b)** $(A \cap B) \cup (A \cap C)$